Context pack: GeminiForge Workbench (the app, not its documents)

Date: 2026-10-21, UPDATED 2026-10-22 (see the last section; newer facts win). Use this as background for every research prompt.

What it is

A documentation workspace that one owner uses, with invited collaborators and AI tools. Structure: Organizations > Projects > Documents. The first project holds a large technical specification ("GeminiForge V20": chapters, a library of crates/packs/prompts, a rule-based question answering engine). Other projects hold ordinary Markdown pages. The planned core product: a master file with phases, steps, sub-steps with check marks, and decisions. Everything is editable by the owner.

Stack and limits

React 18 + TypeScript + Zustand, own dark CSS (no UI library, no Tailwind). Supabase free plan (Postgres with row-level security, Auth, Realtime, Edge Functions, pg_cron). Vercel hobby plan (static site + serverless functions, auto-deploy from GitHub). GitHub Actions CI with Playwright browser tests. No paid services unless named. English only, always: UI, documents and research. Solo owner; today is October 2026.

What exists (v0.7.0)

- Start page = Organizations (cards or list, search, filter, sort, create, rename, soft delete, restore). Organization page: Projects, People, Invitations, Activity, Settings. Project home with documents and a question bar.
- Roles: organization owner / admin / member; project manager / editor / viewer. Members see only projects they are invited to; invitations are single-use links bound to the invited email, hashed in the database, optional email through a serverless function.
- Breadcrumbs, one command palette (Ctrl+K) with scopes and prefixes, collapsible sidebar groups, custom animated dropdowns, compact hover areas.
- AI space: AI tools (through an MCP server in a Supabase Edge Function) read documents and propose edits; every proposal waits in a Review inbox with a diff and risk flags; the owner accepts or rejects; undo; Trash with a timer (default: keep until deleted by hand); tokens with limits, audit log, kill switch.
- Live updates, backups (JSON, llms.txt), import/export of single pages.

Decisions already made

Only signed-in users may see documents (public only through deliberate share links, not built yet). Invited-only access. Keep until deleted by hand is the default for the Trash. Secrets never go into chat, code or git. Documentation content corrections come after the app is finished. Free Supabase plan: no leaked-password protection (TOTP MFA is free per docs/16, to be confirmed in the dashboard).

Open topics (see the research prompts R1-R12)

Editor technology; master-file data model; private publishing and share links; ask-the-docs with verified quotes; MCP security and OAuth; transactional email; free-plan limits; navigation benchmark; accessibility of custom

widgets; import/export; AI-edit review at scale; offline and sync.

Update 2026-10-22 (newer facts win)

Live: PostgreSQL 17.11, database 15 MB of 500 MB, 33 public tables (RLS on), 28 SECURITY DEFINER functions callable by signed-in users, none by anon. Repo is public on GitHub (may go private). Already on Vite 8, TypeScript 7, React 18.3.1; `sb_publishable_`/`sb_secret_` keys already in use. No `vercel.json` (no security headers/CSP yet); sign-in is email + password only (no MFA, no in-app account creation yet); CI has no secret scanning/Dependabot/CodeQL. Supabase Free has NO managed backups (verified). Supabase stops auto-granting new tables and functions on 2026-10-30. Decisions: Canvas single-file build dropped; custom dropdowns stay; share links are project-wide (read-only, expiring, revocable). Full details: the state-update document in the architecture file.

Research status (2026-10-22)

Done and audited: A+B (stack and data), C (security and privacy), E+F (operations and email). Still to do: D+G (accessibility, offline, performance, testing), AI and MCP, feature discovery, UX and visual design. Audit files docs/19, 21, 22 correct earlier statements; where documents disagree, docs/20 (state update) and the audits win. Update: D+G (accessibility, offline, performance, testing) is also done and audited (docs/24). Our TypeScript is 7.0.2, so typescript-eslint cannot be used; lint = Oxlint + tsc. Update: the AI and MCP research is also done and audited (docs/25). Primary in-app Q&A = extractive passage finder with exact quote verification; Gemini app + our MCP = secondary, owner-accepted consumer-cloud path. Remaining research: feature discovery, UX and visual design.